

Aligned but Apart: Why Cosine Conflict Diagnostics Misread

Auxiliary-Loss Interference at the Output Projection

Anonymous authors

Paper under double-blind review

Abstract

Cosine similarity between task gradients is the standard instrument for detecting *conflict* in multi-task and auxiliary-loss training: a negative cosine is read as interference, and gradient-surgery methods act on that signal. We show that at a language model’s shared, tied output projection (the regime this paper is scoped to, distinct from the separate-head multi-token prediction used at scale) this instrument is misleading. Instrumenting the per-row gradients that a next-token (CE) loss and an auxiliary multi-token (MTP) loss deposit on the output projection, we find they are near-perfectly *aligned in direction*: the median per-row cosine is $\approx +1$, only $\approx 0.3\%$ of rows show directional opposition ($\cos < 0$), and 98.4% of rows exceed $|\cos| > 0.3$, against a calibrated control (CE versus an unrelated L1 loss) on the same threshold that reads $\approx 0.4\%$. Yet the *aggregate* cosine over the flattened matrix sits near 0, and we show this decoupling *emerges during training* from an initial state where both metrics agree at $\approx +0.99$. The aggregate near-zero is therefore not the cancellation of opposing per-row gradients; it is consistent with the two losses concentrating their gradient *magnitude* on *different output rows* (*support divergence*; inferred from aligned per-row directions plus a near-zero aggregate, not directly measured; see Section 5). Adding the shared MTP objective degrades next-token loss substantially (+0.39 nats at 124M, Welch’s t , $p \approx 8 \times 10^{-5}$), but this degradation co-occurs with aligned, not conflicting, gradients. We further report that the high per-row alignment is *general* across the model’s weight matrices, persists when the head is *untied*, and that rarer tokens show *higher* per-row alignment than common ones; and that no capacity-separated auxiliary we tried recovers the CE-only baseline (the closest, a tuned contrastive variant, is statistically indistinguishable from baseline at $n=3$, $p \approx 0.19$, but its point estimate still sits above it: non-recovery, not a demonstrated recovery). The methodological consequence is direct: in this regime, cosine-based conflict diagnostics measure the wrong axis, because the operative axis is gradient *support*, not direction. Our account complements Godey & Artzi (2026), who characterize how the output head suppresses a *single* loss’s gradient *norm*; we characterize, in direction-space, what a cosine diagnostic sees when *two* losses share it.

1 Introduction

Predicting several future tokens at once (multi-token prediction, MTP) has become a common ingredient in large-scale language modeling, used to improve sample efficiency and to enable self-speculative decoding (Gloeckle et al., 2024; DeepSeek-AI, 2024). The auxiliary objective is double-edged: a growing body of work reports that adding it can *degrade* the primary next-token objective, particularly at smaller scales and longer horizons (Gloeckle et al., 2024; Zuhri et al., 2025). Asked why, practitioners reach for the intuition that pervades multi-task learning: the two losses must pull the shared parameters in *conflicting directions*, and the remedy is to detect and neutralize that conflict, typically from the sign of a gradient cosine (Yu et al., 2020; Wang et al., 2021). This intuition has been questioned in auxiliary-task learning: Jiang et al. (2023) report that gradient-conflict coordination can overlook auxiliary-target generalization, so the cosine signal

does not by itself determine negative transfer. We ask what that signal actually measures at a language model’s shared output head.

This paper is a measurement study of that intuition for the case of an auxiliary multi-token objective sharing a model’s tied output projection. Our central finding is diagnostic: **the cosine conflict signal is misleading here**. We instrument the gradient each loss deposits on the output projection and examine it **row by row** (one row per vocabulary item). Two facts emerge, both from data already in hand. First, the per-row gradients are not opposed; they are *aligned*: the median per-row cosine sits at $\approx +1$ across the measured window (the first 1000 steps of a dedicated 124M diagnostic run, and out to 87k steps at 350M; Section 4.1), and $\approx 0.3\%$ of rows show negative cosine, while 98.4% exceed $|\cos| > 0.3$ against a control (CE versus an unrelated regularizer) that reads $\approx 0.4\%$ on the same threshold. Second, the *aggregate* cosine over the flattened matrix is nonetheless near zero, and tracking it over training shows the two metrics **start together** (both $\approx +0.99$ at initialization) and then **decouple**: within ~ 200 steps the aggregate falls below zero while the per-row median stays at $\approx +1$. The natural reading of an aggregate near zero (that opposing per-row gradients cancel) cannot hold here, because there is almost nothing to cancel. A cancellation-free way to obtain a near-zero aggregate while per-row directions agree is for the two losses to place their gradient *magnitude* on *different rows*: direction agrees, support diverges.

We frame this **support divergence** as the axis a cosine diagnostic cannot see (Figure 1), and we relate it to the observed degradation without claiming to have isolated its cause. Adding the shared MTP objective degrades next-token loss by ≈ 0.39 nats ($\approx 10\%$) at 124M, a large and unambiguous effect (Welch’s t , $p \approx 8 \times 10^{-5}$), yet it co-occurs with aligned, not conflicting, per-row gradients (the degradation is measured on the full 30,517-step run; per-row alignment on the diagnostic runs above). Three further observations, all from committed runs, are consistent with a capacity/support account rather than a directional-conflict one. (i) Conflict-targeted gradient surgery does not recover the baseline, as expected when there is negligible directional conflict to remove (Appendix A). (ii) Detaching the auxiliary’s gradient from the shared head (stop-gradient) makes the damage *worse*, not better. (iii) Giving the auxiliary its own capacity (a layer-split contrastive auxiliary) reduces but does *not* eliminate the gap: neither the standard nor a hyperparameter-tuned variant recovers the CE-only baseline within noise when aggregated across seeds.

These findings carry a methodological point beyond MTP. Cosine-based conflict detection is a workhorse of multi-task and auxiliary-learning optimization (Yu et al., 2020; Wang et al., 2021; Du et al., 2018), and it is the wrong instrument in this regime: it reports no conflict, yet interference is real. The decomposition that exposes the problem separates **direction** (cosine) from **support** (where each gradient’s magnitude lives). Two additional measurements sharpen the picture and guard against over-reading it. The high per-row alignment is **general** across the network’s weight matrices (attention and MLP matrices show it too, many with strongly *negative* aggregate cosine), so the output head is not special because the metric singles it out; it is consequential because it directly produces the logits. And the alignment **persists when the head is untied**, so it is a property of the output-projection gradient, not an artifact of weight tying. We also observe that *rarer* tokens exhibit *higher* per-row alignment than common ones, under both optimizers. Finally, our account is a direction-space, multi-loss complement to Godey & Artzi (2026), who show that backpropagating a *single* loss through the low-rank output head destroys most of its gradient *norm*; we ask what a *cosine* diagnostic reports when *two* losses share that head, and answer that their per-row directions align while their support appears to compete: an inference from aligned directions plus a near-zero aggregate, which a direct per-row-norm measurement would confirm.

We are explicit about scope. Every result is for a **single shared tied output head**, at 124M ($n=5$) and 350M (single seed), undertrained, small-batch. We do *not* claim that multi-token prediction *as practiced* (separate per-horizon heads) degrades next-token modeling; establishing that would require a separate-head experiment we identify as the primary future step (Section 5), along with $n=3$ at 350M and matched-seed surgery. What we do establish, on data in hand, is that the standard cosine conflict diagnostic misreads this regime, and why.

Contributions. All on a controlled, reproducible GPT-2 testbed (124M primary, 350M illustrative; FineWeb) with open code and a committed statistics script:

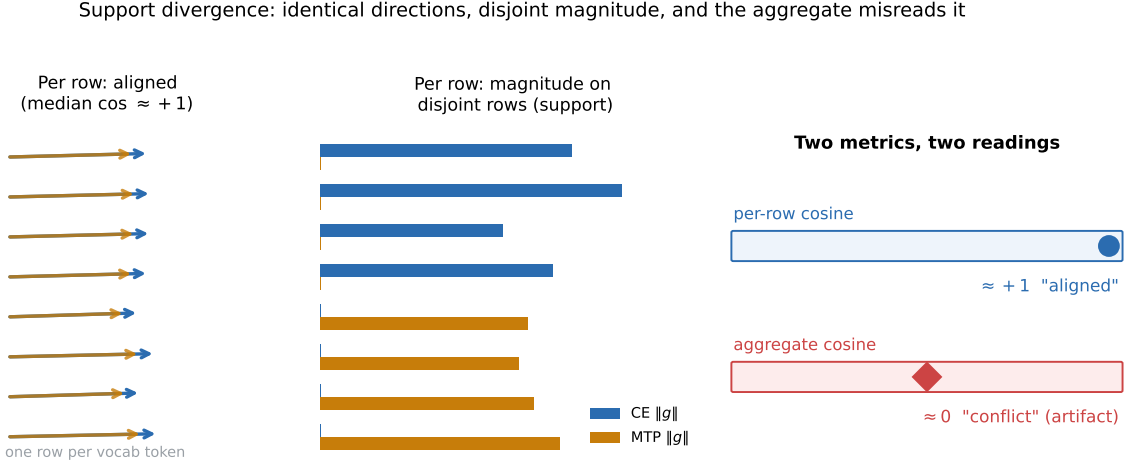


Figure 1: **Support divergence in one picture.** *Left:* row by row (one row per vocabulary token), the CE and MTP gradients point almost the same way, so the per-row cosine is $\approx +1$. *Middle:* but each loss deposits its gradient *magnitude* on a largely disjoint set of rows. *Right:* the per-row cosine correctly reports alignment, while the magnitude-weighted aggregate cosine, the standard conflict diagnostic, reads ≈ 0 and is misinterpreted as conflict. The reading on the right is an artifact of measuring the wrong axis.

- **A diagnostic result (Figures 2, 3).** For an auxiliary multi-token objective sharing the tied output projection, the per-row gradients of the two losses are near-perfectly *aligned in direction* (median cosine $\approx +1$; $\approx 0.3\%$ of rows opposed, against a 0.42% calibrated control) while the aggregate cosine is ≈ 0 , and this decoupling *emerges during training* from an aligned start. Standard cosine-based conflict diagnostics therefore *misdiagnose* this regime.
- **A norm decomposition of the collapse (Figure 4).** Via an exact identity, we show the aggregate would read $+0.94/+0.96$ if gradient magnitude were flat across rows but reads $-0.08/-0.28$: the collapse is set by per-row *norm* structure, not by the typical per-row direction. We ship the instrument (`measure_norm_support.py`) that logs per-row norms to complete the support-versus-opposed-minority test in one short pass.
- **A calibrated, general characterization (Figures 5, 7).** The alignment is **general across the network** and **survives untying** the head, and rarer tokens align *more*. The next-token degradation is large and significant ($+0.39$ nats, $p \approx 8 \times 10^{-5}$) yet co-occurs with aligned gradients.
- **Intervention evidence for a support/capacity account (Figure 6, Appendix A).** Conflict-targeted surgery does not recover the baseline; stop-gradient makes the damage worse; and a capacity-separated auxiliary reduces but does not eliminate the gap (neither standard nor tuned recovers within noise), each reported with its statistical test and caveats.
- **A bridge to the output-layer bottleneck literature.** We position the result as the direction-space, multi-loss complement to Godey & Artzi (2026)’s single-loss, norm-space account of the output head as a gradient bottleneck.

2 Related Work

Multi-token prediction. Predicting several future tokens jointly dates to blockwise-parallel decoding and ProphetNet (Qi et al., 2020), and was revived as a pretraining objective by Gloeckle et al. (2024), who attach n *independent output heads* to a shared trunk and report improved sample efficiency and up to $3\times$ faster inference via self-speculative decoding. Two properties of their result are central to ours: the heads

are *separate* (each future offset gets its own unembedding), and the benefit is *increasingly useful for larger model sizes*. DeepSeek-V3 (DeepSeek-AI, 2024) adopts sequential MTP modules at frontier scale; Medusa (Cai et al., 2024) and related work add prediction heads primarily for speculative decoding (Samragh et al., 2025); and MuToR (Gerontopoulos et al., 2025) interleaves learnable register tokens so a frozen backbone can predict future offsets without architectural change. Across this line, the auxiliary signal is given *its own capacity*: separate heads, modules, or register tokens. The present work studies the opposite, minimal case, a single tied output embedding shared across horizons, precisely because it isolates the interference that separate-capacity designs avoid; a separate-head variant would localize whether the effect is intrinsic to head-sharing (we identify this as the primary future experiment).

Small-scale degradation of MTP. Zuhri et al. (2025) document that multi-token objectives can *hurt* next-token modeling at smaller scales, and propose token-order prediction as an alternative. This degradation is the failure mode we instrument; our contribution is not to add to the catalogue of when MTP helps or hurts, but to measure *what the gradient geometry actually looks like* at the shared head when it hurts, and to show that the standard conflict diagnostic mischaracterizes it.

Gradient conflict and surgery. A large multi-task literature treats negative inter-task gradient cosine as the signature of interference and projects it away: PCGrad (Yu et al., 2020), Gradient Vaccine (Wang et al., 2021), and gradient-similarity loss weighting (Du et al., 2018), with later refinements (Sun et al., 2023; Shi et al., 2023). These methods operate on *global* (flattened) gradient vectors and target gradient *direction*. A parallel line instead balances gradient *magnitude*: GradNorm (Chen et al., 2018) dynamically tunes per-task gradient norms rather than reprojecting directions. This magnitude-versus-direction distinction is central to our findings, which locate the operative structure in per-row gradient norms rather than in the direction the aggregate cosine reports. Independently, Jiang et al. (2023) argue that optimization-based gradient-coordination methods can overlook auxiliary-target generalization, so that gradient-conflict signals do not by themselves determine negative transfer; our measurement gives a concrete mechanism for how a conflict diagnostic can mislead at the output head. Concretely, our per-row measurement shows the global cosine is a lossy summary there: it reads ≈ 0 while the per-row structure is near-perfectly aligned. Conflict surgery therefore has little of the targeted directional conflict to act on (Appendix A), not because surgery is poorly implemented, but because the aggregate is set by per-row norm structure, not by a directional conflict surgery can reproject.

The output head as a bottleneck. Yang et al. (2018) showed the softmax output layer is a rank bottleneck on the distributions a model can express. Godey & Artzi (2026) recently showed, for a *single* loss, that backpropagating through the low-rank output head destroys most of the gradient’s norm, a norm-space single-loss bottleneck. Our result is the direction-space, *multi-loss* complement: we ask what a cosine diagnostic reports when two losses share that head, and find their per-row directions align while their support appears to compete (an inference, not a direct measurement; see Section 5).

Positioning. Relative to prior work, (i) we provide a per-row gradient measurement at the tied output projection that a global cosine misses; (ii) we show the next-token degradation of small-scale shared-head multi-token training, a documented but unexplained failure mode (Zuhri et al., 2025), co-occurs with *aligned* per-row gradients rather than with directional conflict, which we read as *support* competition (inferred, not directly measured); and (iii) our capacity-separated auxiliary *reduces* the next-token cost in the same spirit as capacity-added fixes, though in our runs it does not fully recover the CE-only baseline within noise (neither the standard nor a tuned variant); our contribution is not a fix but the *diagnostic mechanism*: why conflict-based remedies have nothing to act on here, and why capacity separation is the lever that helps.

3 Method

3.1 Setup

We train GPT-2 (124M; 12 layers, $d=768$, vocab 50,257 padded to 50,304) and a 350M variant (24 layers, $d=1024$) on FineWeb, tokenized with the GPT-2 BPE. The output projection (`lm_head`) is *tied* to the input

embedding by default; Section 4.3 reports an untied control. The primary objective is next-token cross-entropy (CE). The auxiliary multi-token objective (MTP) supervises the *same tied head* to also predict tokens at offsets $t+2$ and $t+3$, with total loss $\text{CE} + 0.5 \text{MTP}_2 + 0.25 \text{MTP}_3$. Evaluation is **pure next-token CE on held-out validation** for every variant, so all reported losses are directly comparable. The loss-comparison runs use a batch of 16,384 tokens/step for 30,517 steps ($\approx 500\text{M}$ tokens, ≈ 4.0 tokens/parameter at 124M), an intentionally undertrained, small-batch regime, chosen for controlled comparison and because it is where shared-head multi-token training is already documented to underperform; we return to this scope in Section 5.

3.2 The per-row instrument

The output projection has shape (V, d) ; row i is the weight vector for vocabulary token i . For a validation batch we compute the gradient each loss deposits on this matrix and, for **each row**, the cosine between the CE and MTP row-gradients. We summarize the resulting distribution of V per-row cosines by its median, the fraction with $\cos < 0$ (directional opposition), and the fraction with $|\cos| > 0.3$ (appreciable alignment). We contrast these per-row statistics with the **aggregate** cosine (the cosine between the two *flattened* matrix gradients), which is the quantity conventional conflict diagnostics use. The 47 padding rows (50,304–50,257) are masked from all per-row statistics.

Measurement windows (stated explicitly). The gradient instrumentation and the loss comparison come from *different* runs, and we are careful to keep them distinct. At 124M, the per-row/aggregate trajectory is a dedicated diagnostic run logged densely over its first 1000 steps (A1); the +0.39-nat loss gap is the separate 30,517-step training comparison (Section 4.4). Late-training persistence of the alignment is therefore *not* established at 124M; it is measured directly only at 350M (single seed), where per-row $|\cos| > 0.3$ remains high out to step 87,000 (Section 4.5). We flag throughout which quantity comes from which run, and treat single-seed/single-run measurements as such.

3.3 Direction versus support

The distinction the paper turns on: the **aggregate** cosine measures whether the summed gradient vectors point the same way; the **per-row** distribution measures whether they agree row by row. These can disagree. If two losses each place their gradient magnitude on a *disjoint* set of rows, then on the rows where either is active the other is ≈ 0 , the per-row cosines agree where they overlap, and the flattened aggregate is driven toward orthogonality by the non-overlap, *without any directional opposition*. We call this **support divergence**, and it is invisible to a cosine conflict diagnostic by construction.

A two-row illustration. Take a weight matrix with two rows and let the CE and MTP gradients be $g^{\text{ce}} = [(1, 0), (0, 0)]$ and $g^{\text{mtp}} = [(0, 0), (1, 0)]$: each loss loads all its magnitude on a *different* row. The per-row cosine is undefined on the zero rows and otherwise never negative, so nothing looks opposed. Yet the flattened vectors $(1, 0, 0, 0)$ and $(0, 0, 1, 0)$ are orthogonal: the **aggregate cosine is 0**. No row’s gradients point in opposing directions; the aggregate reads “conflict” purely because the two losses occupy disjoint rows. Support divergence is this effect at scale.

One caveat bounds how far we can push this inference. The aggregate cosine is *magnitude-weighted* (it is the cosine of the summed, flattened gradients), whereas the per-row median and the $\cos < 0$ fraction are *unweighted* over rows. A near-zero aggregate with a per-row median of +1 is therefore consistent with (a) disjoint magnitude support across rows, but also with (b) a small number of high-norm rows being off-axis while the many low-norm rows align, i.e. the 0.33% opposed-by-count figure does not bound the opposed *mass*. Our committed measurements store per-row cosines but not per-row gradient norms, so we cannot separate (a) from (b) here; we report support divergence as the reading most consistent with the full pattern (including the emergent decoupling and the untied-head control) and flag the direct per-row-norm measurement that would settle it as future work (Section 5).

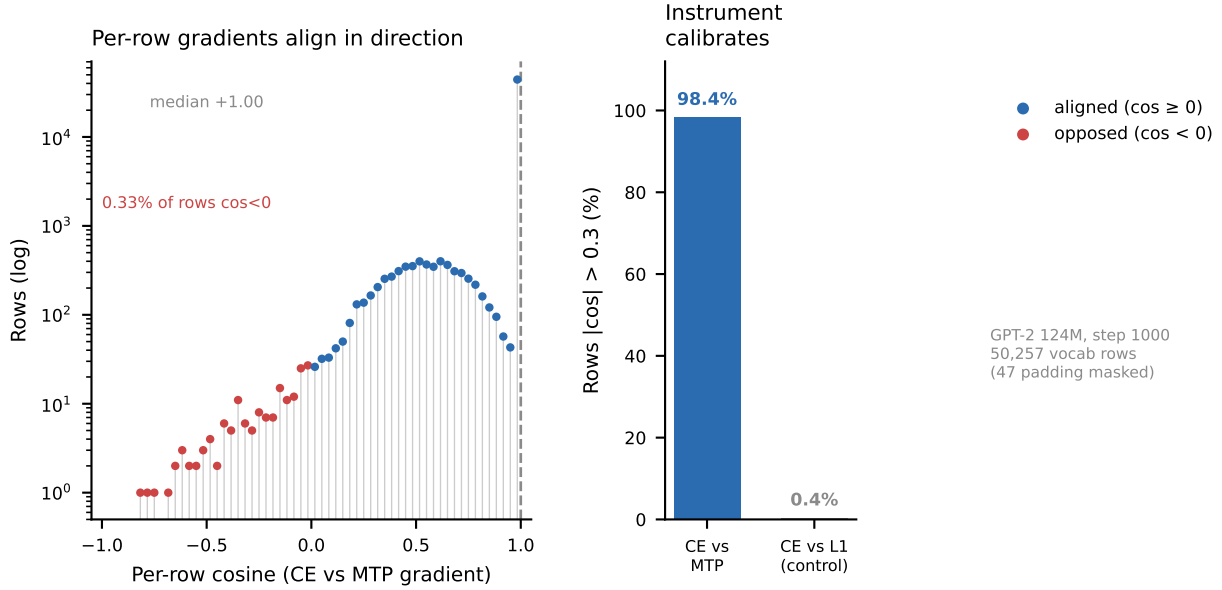


Figure 2: **Per-row CE/MTP gradients on the output projection are near-perfectly aligned.** Left: distribution of per-row cosines (median +1.00; 0.33% opposed, in red). Right: a CE-vs-L1 control reads 0.42% of rows above $|\cos| > 0.3$ versus 98.4% for CE-vs-MTP: the instrument calibrates. GPT-2 124M, step 1000, 50,257 vocab rows (47 padding rows masked).

3.4 Calibration

To confirm the instrument is not trivially reporting alignment, we run a control (A3) that measures CE against an unrelated L1 regularizer on the same head. A well-calibrated instrument should read near-zero alignment for two unrelated losses.

4 Experiments and Results

4.1 Per-row gradients are aligned, not conflicting

At 124M, the per-row CE/MTP gradients on the output projection are near-perfectly aligned: **median per-row cosine** +1.00, with only **0.33% of rows** showing $\cos < 0$ and **98.4%** exceeding $|\cos| > 0.3$ (step 1000, 50,257 rows). The control reads **0.42%** above the same threshold, two orders of magnitude lower, confirming the instrument distinguishes aligned from unrelated losses (Figure 2). There is essentially no directional conflict to detect.

4.2 The aggregate cosine decouples from per-row alignment during training

The aggregate cosine nonetheless reads ≈ 0 (-0.08 at step 1000). Tracking both metrics over training (Figure 3) shows this is not a fixed property: at initialization the aggregate and per-row alignment **agree at** $\approx +0.99$, and the aggregate then crosses zero early (between steps 100 and 150 under Muon, and between 150 and 200 under AdamW) while the per-row median holds at $\approx +1$ for the remainder of the 1000-step diagnostic window (late-training persistence is shown separately at 350M, Section 4.5). These trajectories are a single diagnostic run per optimizer; the close agreement between two different optimizers is the replication we have.

We can pin down *how* the aggregate collapses without any additional runs, using an exact identity. The aggregate cosine is a magnitude-weighted average of the per-row cosines, $\text{agg} = \sum_i w_i \cos_i$ with weights

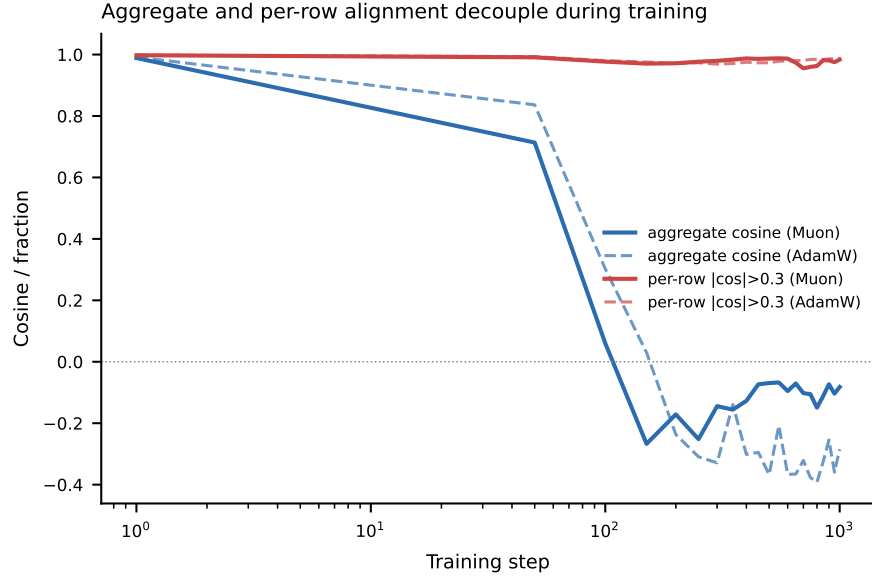


Figure 3: **The aggregate and per-row alignment start together and decouple.** Both metrics begin at $\approx +0.99$; within ~ 200 steps the aggregate (matrix-level) cosine falls below zero (crossing between steps 100–150 for Muon and 150–200 for AdamW) while per-row alignment stays ≈ 0.98 . One diagnostic run per optimizer (Muon solid, AdamW dashed); the agreement between two optimizers is the replication.

$w_i = \|g_i^{\text{ce}}\| \|g_i^{\text{mtp}}\| / (\|G^{\text{ce}}\| \|G^{\text{mtp}}\|) \geq 0$. If gradient magnitude were spread evenly across rows, the weights would be equal and the aggregate would equal the *plain mean* of the per-row cosines. That mean is $+0.94$ (Muon) and $+0.96$ (AdamW), whereas the actual aggregate is -0.08 and -0.28 (Figure 4). The entire ≈ 1.0 gap between “aggregate if norms were flat” and the real aggregate is therefore attributable to the per-row *norm* structure, not to the typical per-row direction: the standard diagnostic is reporting a magnitude-weighting artifact, which is the paper’s central point made quantitative.

What the identity does *not* settle is which norm structure is responsible. A negative aggregate requires the opposed minority ($\cos < 0$, $\approx 0.33\%$ of rows) to carry *some* weight, so two mechanisms remain compatible with the data: (a) *support divergence*, where CE and MTP load their magnitude onto increasingly disjoint rows, making the products w_i small precisely where each gradient is large; and (b) a small number of *high-norm opposed rows* dominating the weighted sum. Separating (a) from (b) needs the per-row gradient *norms*, which the committed runs do not store. We provide a drop-in instrument (`measure_norm_support.py`) that logs them and reports the two disambiguating statistics (norm-profile overlap and opposed-norm fraction) on the same short diagnostic run; running it is the single measurement that would convert support divergence from the reading most consistent with the evidence into a direct result (Section 5).

4.3 The pattern is general, and survives untying

High per-row alignment is **not unique to the output head**. Across all 74 weight matrices (Figure 5), attention and MLP matrices also show high per-row alignment (many above 0.8), frequently with strongly *negative* aggregate cosine (down to -0.78). The output head is distinctive only in being the point of **maximum** per-row alignment (0.984) sitting nearest aggregate-zero, and, more importantly, in being the matrix that *directly produces the logits*. The head matters not because the metric singles it out, but because interference there is paid directly in next-token loss. Alignment also **persists when the head is untied** (A4: 99.3% of rows above threshold, versus 98.7% tied at the matched step), so it is a property of the output-projection gradient ($h^\top \partial \mathcal{L} / \partial \text{logits}$), not an artifact of weight tying. The per-layer generality (Figure 5) and the untied-head comparison are each *single-run* measurements at a fixed step; we report them as point observations, not seed-replicated estimates (only the 124M CE-vs-MTP loss gap is replicated across seeds).

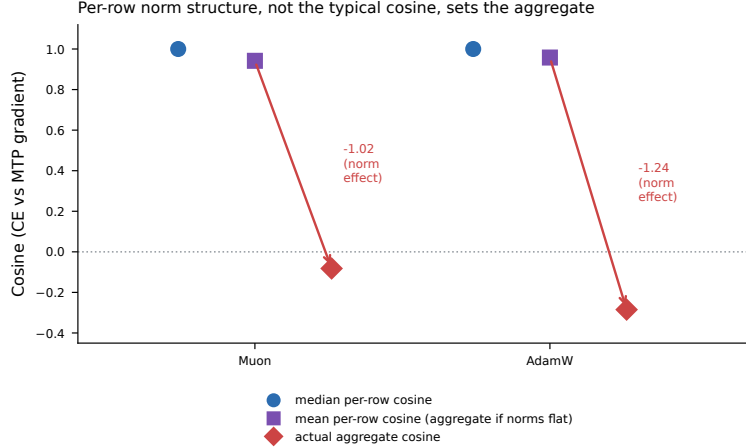


Figure 4: **Per-row norm structure, not the typical cosine, sets the aggregate.** For each optimizer: the median per-row cosine (+1.00), the plain *mean* per-row cosine (which equals the aggregate the standard diagnostic would report if gradient magnitude were spread evenly across rows), and the *actual* aggregate cosine. The mean is strongly positive (+0.94/+0.96) while the actual aggregate is negative (−0.08/−0.28); the gap is due entirely to per-row norm weighting. Single diagnostic run per optimizer, step 1000, 50,257 real vocabulary rows.

We are deliberate about what this generality does and does not license. The same qualitative signature (high per-row alignment with a non-positive aggregate) appears across the network, and at many matrices the aggregate is *strongly* negative (to −0.78), not merely near zero. This means the per-row/aggregate gap is a fairly generic property of these structured gradient matrices, so a near-zero aggregate on its own is *not* a specific signature of two-loss support competition. What is specific to the output head is (a) its aggregate is close to zero rather than strongly negative, which is the regime in which the “no cancellation” reading is cleanest (there is little opposing mass to cancel), and (b) it is the one matrix whose per-row gradients are the CE and MTP gradients directly, and whose interference is paid immediately in next-token loss. We do not adjudicate the strongly-negative aggregates elsewhere; a per-row account of those is beyond what the committed measurements settle.

4.4 Adding shared MTP degrades next-token loss; no auxiliary recovers it

Adding the shared MTP objective degrades next-token validation loss from 3.985 ± 0.014 (CE-only, $n=5$) to 4.377 ± 0.021 ($n=3$): a +0.39 nat ($\approx 10\%$) increase, Welch’s $t = -28.8$, $p \approx 7.6 \times 10^{-5}$, Cohen’s $d = -23.7$; the very large d reflects the tight seed-to-seed reproducibility of this controlled testbed (within-condition sd ≈ 0.01 – 0.02) rather than an unusually large real-world effect beyond the 0.39-nat gap itself; the 95% CI on the difference (Welch) is [0.35, 0.43] nats. Detaching the auxiliary from the head via stop-gradient makes it *worse* (4.493, $\Delta+0.508$, $p \approx 5.5 \times 10^{-9}$), not better, which points to a support/capacity account rather than directional conflict (we note this is a data point, not a proof: stop-gradient also removes the auxiliary’s ability to shape the shared representation, which is a competing reason it could hurt). The $n=5$ CE-only and $n=3$ shared-MTP conditions share three seeds (42, 123, 456); the two extra CE-only seeds (789, 1337) were run for a tighter baseline and no MTP seeds were dropped. Capacity-separated auxiliaries reduce but do **not** recover the baseline. The *standard* layer-split contrastive auxiliary reaches 4.024 ($\Delta+0.039$, $p \approx 0.010$), statistically distinguishable from A, i.e. a genuine non-recovery. A hyperparameter-tuned variant reaches 4.003 ($\Delta+0.018$) but is *not* statistically separable from A at $n=3$ ($p \approx 0.19$): for this variant we can claim neither recovery nor non-recovery, and we rest the capacity conclusion on the standard variant. A latent-prediction baseline (NextLat) lands at 4.037 ($\Delta+0.051$) (Figure 6).

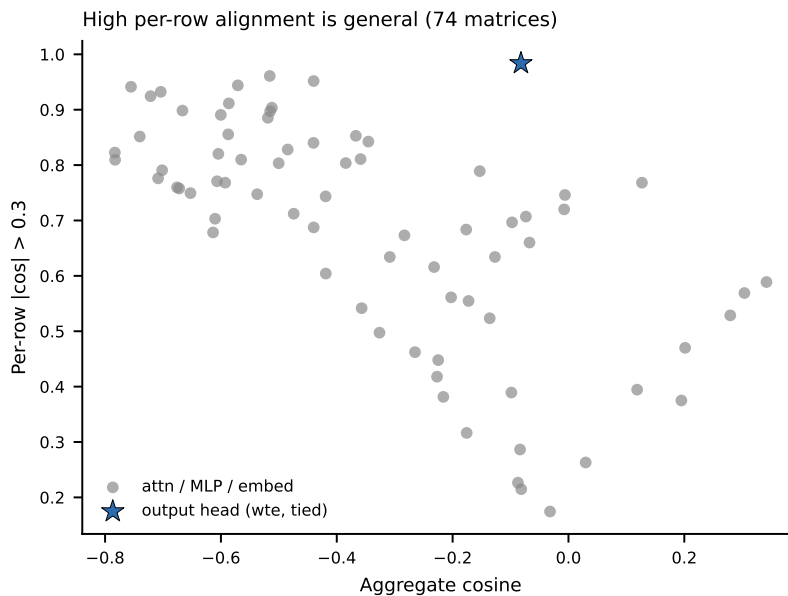


Figure 5: **High per-row alignment is general across the network.** Per-row alignment vs aggregate cosine for all 74 weight matrices. The output head (star) is the maximum and sits nearest aggregate-zero, but attention/MLP matrices show the same pattern, many with strongly negative aggregate cosine.

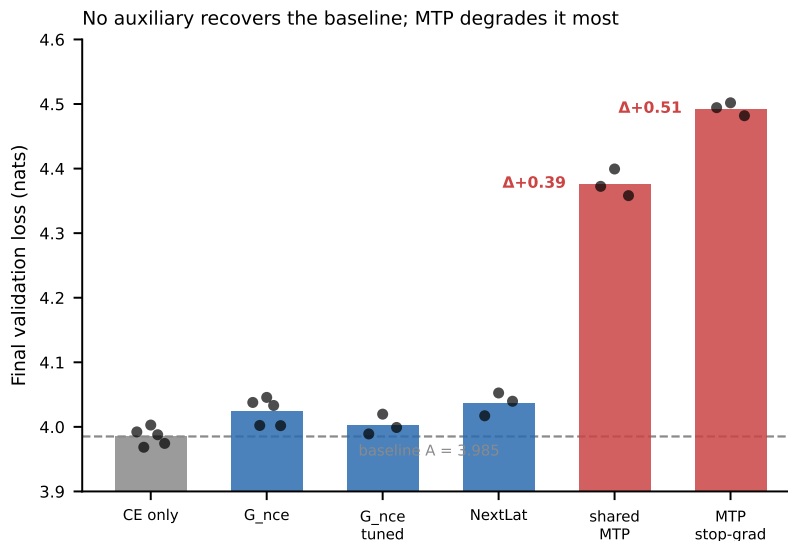


Figure 6: **No auxiliary recovers the baseline; MTP degrades it most.** Final next-token validation loss by variant (points = seeds). Shared MTP degrades CE by $\Delta+0.39$; stop-grad by $\Delta+0.51$; no capacity-separated auxiliary recovers the CE-only baseline within noise.

That conflict-targeted gradient surgery also does not recover the baseline is as expected when there is negligible directional conflict to remove; because those runs are single-seed with a numerical-drift caveat, we report them in full in Appendix A rather than resting any claim on them.

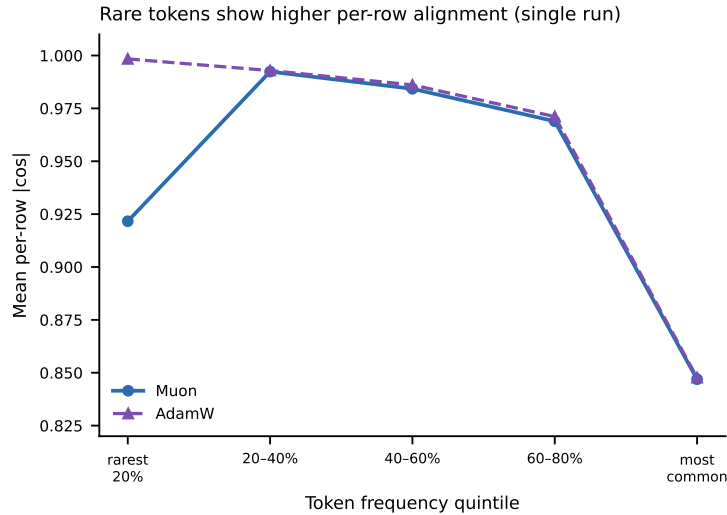


Figure 7: **Rare tokens show higher per-row alignment.** Mean per-row $|\cos|$ by token-frequency quintile (a non-saturating statistic; the $|\cos| > 0.3$ fraction is near-ceiling for all bins). Rarer tokens align more than common ones under both optimizers. Single run per optimizer.

4.5 Rare tokens align more; the pattern holds at 350M

Per-row alignment is somewhat *higher* for rarer tokens. We report this on the *mean* per-row $|\cos|$ per frequency quintile (rather than the $|\cos| > 0.3$ fraction, which is near-saturated for all bins and has little headroom): the rarest quintile averages 0.922 (Muon) / 0.998 (AdamW) versus 0.847 / 0.848 for the most-common quintile (Figure 7). This is a single-run measurement at a fixed step; we present it as suggestive, and its value is that the ordering is consistent across two optimizers. The direction-vs-support picture also holds qualitatively at **350M** (single seed): CE-only 3.752, shared-MTP 4.129 ($\Delta+0.378$).

The 350M runs are the paper’s only *late-training* per-row measurement, logged out to step 87,000. The instrument computes the same quantity in both runs (the per-row cosine between the CE gradient and the MTP gradient on the output projection) regardless of which objective the model was trained with (the MTP gradient is computed at measurement time for both). Measured this way, the shared-MTP-trained model holds $|\cos| > 0.3$ at 98.9% of rows at step 87,000, and the CE-only-trained model reads 94.5% at the same step. That the CE-only model *also* shows high CE/MTP alignment this late indicates the alignment is an intrinsic property of the output-projection gradient geometry, not something the MTP objective induces. Both are single seeds, so we treat the persistence as illustrative (consistent with alignment not decaying over a long horizon), not established.

5 Discussion

What the measurements establish. In this regime (a single shared tied output projection, $\leq 350M$, undertrained) the standard cosine conflict diagnostic misreads the CE/MTP interaction. The per-row gradients are aligned (median +1), the aggregate is ≈ 0 , and the two decouple during training in a pattern that points to support divergence rather than cancellation. The next-token degradation is large and unambiguous, yet it coincides with aligned gradients, so conflict-based remedies have essentially nothing to act on, consistent with the surgery baselines in Appendix A, which recover nothing. The lever that helps is capacity separation, though in our runs it reduces rather than eliminates the gap.

What they do not establish. We do not show that MTP *as practiced* (with separate per-horizon heads) degrades next-token modeling; our formulation is a single tied head, and separate-head designs give the auxiliary its own capacity precisely so that support cannot compete. We do not establish causation between

support divergence and the loss increase; we show they co-occur and that interventions behave as a support-/capacity account predicts. And the effect lives in an undertrained, small-batch, $\leq 350\text{M}$ regime where MTP is already documented to underperform (Zuhri et al., 2025).

Future work. Four experiments would broaden the scope from “the diagnostic misreads this regime” toward a causal, general statement: (1) a **separate-head MTP** variant, to localize whether the degradation is intrinsic to head-sharing; (2) $n=3$ **at 350M** plus a 124M CE-only per-row baseline, to turn the single-seed scale observation into a measured trend; (3) **matched-seed, single-pass gradient surgery** at $n=3$, to remove the numerical-drift caveat; and (4) a **direct per-row-norm measurement**, to upgrade support divergence from an inference (forced by aligned per-row + zero aggregate) to a measurement. For (4) we have already shown, via an exact identity, that the aggregate collapse is driven by per-row norm structure rather than by the typical per-row direction (Section 4.2, Figure 4), and we ship the instrument (`measure_norm_support.py`) that logs the per-row norms and reports the two statistics (norm-profile overlap, opposed-norm fraction) needed to separate support divergence from a high-norm opposed minority; it runs on the existing short diagnostic pass, so (4) is a single measurement rather than a new training campaign.

6 Conclusion

Cosine similarity between task gradients is the default instrument for detecting interference in multi-task and auxiliary-loss training. We showed that at a language model’s shared, tied output projection this instrument is misleading: the per-row CE and MTP gradients are near-perfectly aligned in direction, the aggregate cosine is driven to zero by disjoint support rather than by cancellation, and the two metrics decouple during training. Adding the shared auxiliary degrades next-token loss substantially, but not through the directional conflict a cosine diagnostic would report. The operative axis is gradient support, not direction, a distinction that explains why conflict-targeted surgery does not help here and why capacity separation does. We deliver the result on a fully reproducible testbed with a committed statistics script, and we are explicit about the shared-head, small-scale scope and the experiments that would extend it.

Reproducibility Statement

All results are computed from committed result JSONs by `analysis/stats.py`; the six data figures are regenerated from the same JSONs by `figures/make_figures.py` (the conceptual schematic, Figure 1, by `figures/make_schematic.py`). Code, data-loading, pinned dependencies (`requirements.txt`), and a README with exact run commands are released. The optimizer (Muon) is pinned to a specific commit. Scope and caveats are stated in Section 1 and Section 5.

References

- Tianle Cai, Yuhong Li, Zhengyang Geng, Hongwu Peng, Jason D. Lee, Deming Chen, and Tri Dao. Medusa: Simple llm inference acceleration framework with multiple decoding heads. *arXiv preprint arXiv:2401.10774*, 2024.
- Zhao Chen, Vijay Badrinarayanan, Chen-Yu Lee, and Andrew Rabinovich. GradNorm: Gradient normalization for adaptive loss balancing in deep multitask networks. In *International Conference on Machine Learning (ICML)*, 2018.
- DeepSeek-AI. Deepseek-v3 technical report. *arXiv preprint arXiv:2412.19437*, 2024.
- Yunshu Du, Wojciech M. Czarnecki, Siddhant M. Jayakumar, Mehrdad Farajtabar, Razvan Pascanu, and Balaji Lakshminarayanan. Adapting auxiliary losses using gradient similarity. *arXiv preprint arXiv:1812.02224*, 2018.
- Anastasios Gerontopoulos, Dimitrios Metaxas, Feng Yang, et al. Multi-token prediction needs registers. *arXiv preprint arXiv:2505.10518*, 2025.

- Fabian Gloeckle, Badr Youbi Idrissi, Baptiste Rozière, David Lopez-Paz, and Gabriel Synnaeve. Better & faster large language models via multi-token prediction. *arXiv preprint arXiv:2404.19737*, 2024.
- Nathan Godey and Yoav Artzi. Lost in backpropagation: The lm head is a gradient bottleneck. *arXiv preprint arXiv:2603.10145*, 2026.
- Junguang Jiang, Baixu Chen, Junwei Pan, Ximei Wang, Dapeng Liu, Jie Jiang, and Mingsheng Long. ForkMerge: Mitigating negative transfer in auxiliary-task learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- Weizhen Qi, Yu Yan, Yeyun Gong, Dayiheng Liu, Nan Duan, Jiusheng Chen, Ruofei Zhang, and Ming Zhou. Prophetnet: Predicting future n-gram for sequence-to-sequence pre-training. *arXiv preprint arXiv:2001.04063*, 2020.
- Mohammad Samragh et al. Your llm knows the future: Uncovering its multi-token prediction potential. *arXiv preprint arXiv:2507.11851*, 2025.
- Guangyuan Shi et al. Recon: Reducing conflicting gradients from the root for multi-task learning. *arXiv preprint arXiv:2302.11289*, 2023.
- Yi Sun et al. Towards multi-objective optimization with dynamic reweighting of multiple gradients. *arXiv preprint arXiv:2305.19844*, 2023.
- Zirui Wang, Yulia Tsvetkov, Orhan Firat, and Yuan Cao. Gradient vaccine: Investigating and improving multi-task optimization in massively multilingual models. In *International Conference on Learning Representations (ICLR)*, 2021.
- Zhilin Yang, Zihang Dai, Ruslan Salakhutdinov, and William W. Cohen. Breaking the softmax bottleneck: A high-rank rnn language model. In *International Conference on Learning Representations (ICLR)*, 2018.
- Tianhe Yu, Saurabh Kumar, Abhishek Gupta, Sergey Levine, Karol Hausman, and Chelsea Finn. Gradient surgery for multi-task learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.
- Zayd M. K. Zuhri et al. Predicting the order of upcoming tokens improves language modeling. *arXiv preprint arXiv:2508.19228*, 2025.

A Gradient-surgery baselines (single-seed; reported for completeness)

We ran conflict-targeted gradient-surgery methods on the shared-MTP variant to test directly whether removing directional conflict recovers the baseline. **We do not rest any claim in the main text on these runs**, for three reasons stated up front: (i) each is a **single seed** (seed 42); (ii) the PCGrad baseline operates on **all parameters**, whereas per-row Gram-Schmidt and Scatter operate on the **output head only**, so they are not strictly comparable; and (iii) surgery uses **two backward passes** under bf16 (versus the fused single pass of the main runs), introducing a $\sim 0.02\text{--}0.04$ nat numerical drift comparable to the smaller surgery effects. With those caveats, the pattern is uniform: no surgery method recovers the CE-only baseline ($A = 3.985$), and none improves meaningfully on the matched shared-MTP seed ($B@42 = 4.390$).

This is what a support/capacity account predicts: with per-row directions already aligned, conflict surgery has almost nothing to project away. A matched-seed, single-pass replication at $n=3$ (identified in Section 5) would remove the drift caveat and is the right way to upgrade this from a consistent-with observation to evidence. The AdamW negatives from the same phase ($A = 3.743/3.754$, $B = 4.149/4.153$, $G_{\text{ncc}} = 3.778/3.781$ across seeds 42/123) reproduce the ordering under a second optimizer and are included in `results/phase_c/`.

Method	Acts on	Final val loss	Δ vs B@42	Δ vs A
CE only (A, ref)	—	3.985	—	0.000
shared MTP (B@42)	—	4.390	0.000	+0.405
Per-row Gram-Schmidt	head only	4.418	+0.027	+0.432
Scatter	head only	4.458	+0.067	+0.472
PCGrad	all params	4.544	+0.154	+0.559
Pure-Muon (all 2D in Muon)	all params	4.556	+0.166	+0.571

Table 1: Gradient-surgery baselines on shared MTP (seed 42). Every method leaves the loss at or above the shared-MTP baseline; the per-row Gram-Schmidt effect (+0.027) sits inside the two-pass drift envelope, so we treat it as “no recovery,” not as a measured improvement.